

Omar Ali

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EDUCATION

UC Berkeley, College of Engineering

Berkeley, CA

Bachelor of Science in Electrical Engineering and Computer Sciences (GPA 3.327)

August 2024 - May 2026

- **Relevant Coursework:** The Structure and Interpretation of Computer Programs, Computer Architecture, Software Engineering, Artificial Intelligence, Computer Security, Internet Architecture and Protocols, Database Systems, Circuits & Devices

College of Marin

Kentfield, CA

Associate of Science in Mathematics (GPA 4.0)

June 2021 – May 2024

- **Relevant Coursework:** Data Structures and Algorithms (C++), Advanced Java, Advanced C++, Computer Organization

SKILLS

Languages: Python, IDL, Java, C/C++, Ruby, JavaScript, MATLAB, SQL

Frameworks: TensorFlow, Ruby on Rails, pytest, SciPy, PyQt, Cucumber, RSpec, Pillow, scikit-image

Tools: Git, Jira, Heroku

Spoken Languages: Native English, Conversational Spanish, Elementary Arabic

WORK EXPERIENCE

Lawrence Livermore National Laboratory (LLNL)

Livermore, CA

Software Engineer Intern, National Ignition Facility

May 2024 – May 2026

- Led migration of a 75,000-line legacy IDL codebase to Python, modernizing tooling used by engineers at the National Ignition Facility to analyze laser shot data through automated metrics and visualizations. Released facility-wide in April 2026.
- Developed a comprehensive test suite using pytest, including unit, integration, and UI tests to ensure compatibility between the original tool and the ported application.
- Collaborated with scientists, engineers, and operators to ensure program met user specifications and additional feature requests.

Salisbury University — NSF REU

Salisbury, MD

Undergraduate Researcher; Advisor: Dr. Yuanwei Jin (University of Maryland)

May 2023 – Aug 2023

- Developed a fully convolutional network (TensorFlow) for semantic tumor segmentation on diffusion-weighted MRI, supporting Dr. He's (Purdue) bacteria-based cancer therapy research — achieved 98% segmentation accuracy, 92% precision.
- Preprocessed raw DICOM data in MATLAB/IDL and built an augmentation pipeline in SciPy, Pillow, and scikit-image to address dataset scarcity.
- Implemented class weighting to address imbalance, improving precision by 15%.
- Presented findings at ABRCMS 2023 (award in Computational & Systems Biology).

Computing Talent Initiative

Monterey, CA

Student Accelerator

June 2022 – May 2023

- Tutored coursework focused on data structures and algorithms in Python for a nonprofit, and mentored a cohort of about 30 college students.
- Guided students through open-source contributions in partnership with the organization CodeDay Labs.

PROJECTS

Civic Engagement Web App

- Built a Rails web app with a team of four (deployed to Heroku) that lets users find their U.S. representatives by state, county, or city via an interactive map, using the Congress.gov RESTful API to pull live legislator and bill data.
- Designed and implemented the representative-following feature: a PostgreSQL-backed ratings system letting users save representatives, rate them with a 5-star scale, and receive a personalized feed of their news and legislative activity.
- Practiced BDD/TDD with Cucumber and RSpec in a 4-person team, using Git-based code review to integrate independently developed features.

Memory-Safety Exploitation

- Exploited a series of vulnerable C programs by crafting stack-smashing and buffer-overflow attacks to hijack control flow, reasoning about stack layout, calling conventions, and defenses such as stack canaries and ASLR.